

# 05 Linear Regression Salary

## Markdown Cell 1

Experiment 5 — Linear Regression on Salary Data  
Source: Linear Regression PDF

Implementation of Linear Regression on Salary Data.

## Code Cell 2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('Salary_Data.csv')
df.head(10)
```

## Code Cell 3

```
df.tail()
df.shape
```

## Code Cell 4

```
x=df.iloc[:, :-1]
y=df.iloc[:, 1]

print(x)
print(y)
```

## Code Cell 5

```
from sklearn.model_selection import train_test_split

x_train,x_test,y_train,y_test = train_test_split(
    x,y,test_size=0.2,shuffle=True
)

print(x_train)
print(y_train)
print(x_test)
print(y_test)
```

## Code Cell 6

```
from sklearn.linear_model import LinearRegression

model=LinearRegression()
model.fit(x_train,y_train) # fit() function used for training a model
model.score(x_test,y_test) # score() function used for finding accuracy
```

## Code Cell 7

```
model.coef_ # value of 'm'
model.intercept_ # value of 'c'
```

## Code Cell 8

```
x_test
y_test
```

## Code Cell 9

```
y_pred=model.predict(x_test)
y_pred
```

## Code Cell 10

```
y_predict=model.predict([[22]])
y_predict
```

## Code Cell 11

```
x_pred = model.predict(x_train)
x_pred
```

## Code Cell 12

```
y_test[:6]  
model.predict(x_test[:8])
```

### Code Cell 13

```
plt.scatter(x_train,y_train,color='blue')  
plt.plot(x_train,x_pred,color='pink')  
plt.title('Salary vs Experience (Training Dataset)')  
plt.xlabel('YearsExperience')  
plt.ylabel('Salary')  
plt.show()
```

### Code Cell 14

```
plt.scatter(x_test,y_test,color='green')  
plt.plot(x_train,x_pred,color='blue')  
plt.title("Salary vs Experience (Training Dataset)")  
plt.xlabel("YearsExperience")  
plt.ylabel("Salary(in Ruppeess)")  
plt.show()
```